

Piyush Raj

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ABOUT

Passionate **Software Developer** with **full-stack** and **MERN stack** expertise, building **scalable web platforms** and **data-driven systems**. Experienced in **RESTful APIs**, **system design**, and **AI-powered solutions**. Strong foundation in **software engineering**, **database systems**, and **cloud-native development**, with experience using **Azure**, **Databricks**, and modern **AI development tools** to build reliable, user-centric applications.

TECHNICAL SKILLS

Languages: C++, Python, SQL

AI/ML: Scikit-learn, TensorFlow, PyTorch, NumPy, Pandas, XGBoost, OpenCV, NLP, Transformers, LLMs, RAG, Prompt Engineering, Vector Databases, Feature Engineering, Time Series Analysis

Backend: FastAPI, Flask, RESTful APIs, WebSockets, JWT Auth, RBAC

Databases: PostgreSQL, MongoDB, MS SQL Server, Firebase

Cloud/DevOps: Docker, Git, GitHub, CI/CD, Databricks

Tools: Jupyter Notebook, VS Code, Cursor

Core CS: Data Structures & Algorithms, OOP, DBMS, Computer Networks, Operating Systems, System Design, Design Patterns

AI Platforms: Gemini AI, Claude AI, ChatGPT, LangChain, LangGraph

PROJECT

TravelOS – AI Travel Planning Platform — Python, LangGraph, AI Agents, RAG, Docker [🐙 GitHub](#) [🔗 Live](#)

- **Engineered an autonomous multi-agent travel platform** using **LangGraph**, **Google Gemini**, and **FastAPI**, enabling conversational planning and personalized itineraries.
- **Integrated specialized AI agents** with **FAISS Vector RAG**, multi-currency processing, and **React** to deliver intelligent recommendations and synchronized travel planning.

Smart Personal Expense Analyzer — React, FastAPI, Python, MongoDB, Machine Learning [🐙 GitHub](#) [🔗 Live](#)

- **Built a full-stack AI finance platform** with **React** and **FastAPI** supporting expense management, JWT authentication, analytics, AI categorization, and personalized financial insights.
- **Engineered an intelligent financial analysis pipeline** processing transaction data for **automated categorization**, spending pattern analysis, anomaly detection, and personalized recommendations through modular AI services.

AuthAI – Real-Time Behavioral Biometrics Authentication — Python, Scikit-learn, TensorFlow [🐙 GitHub](#)

- **Architected a real-time behavioral authentication system** using **Python**, **Scikit-learn**, **TensorFlow**, and **MongoDB**, combining behavioral biometrics with secure user authentication and session management.
- **Benchmarked multiple ML/DL models** including **Random Forest**, **XGBoost**, **Isolation Forest**, **LSTM**, and **Transformers** for real-time bot detection using mouse, keyboard, focus, and click behavior.

EDUCATION

Chandigarh University (CU)

B.E. in Computer Science and Engineering — CGPA: 8.26

Mohali, Punjab, India

Aug 2022 – June 2026

Krishna Public School

High School: 85.83% — Intermediate: 74.67%

Bilaspur, Chhattisgarh, India

2020 – 2022

CERTIFICATIONS & ACHIEVEMENTS

Oracle Cloud Infrastructure 2025 Certified AI Foundations Associate

Oracle

SQL: A Practical Introduction for Querying Databases

IBM

Published Research: AuthAI – Real-Time Behavioral AI for CAPTCHA-Free Security

International Journal of Scientific Research in Engineering and Management (IJSREM)

Project Expo – 2nd Place out of 40+ teams

College-Level Project Expo – AI-Based Project